

# RELOC8 — AI & GIS-Based Disaster Management and Intelligent Relocation Ecosystem

## 1. Executive Summary

**RELOC8** is an AI- and GIS-based disaster management and intelligent relocation platform designed to support citizens and authorities before, during, and after disasters.

The system brings together **risk assessment, GIS-based mapping, emergency communication, medical assistance, relocation planning, shelter/resource information, animal assistance, and offline support** in a single platform.

RELOC8 follows a dual-interface approach:

- **Citizen View** – allows citizens to understand local risks, report emergencies, access important information, request medical assistance, and find support.
- **Authority Dashboard** – enables authorities to monitor emergency requests, assess risk information, coordinate response, and manage relocation-related information.

The prototype is designed around the Bihar pilot region, particularly **Supaul and Madhepura**, where flood-related risks make timely evacuation and relocation decisions especially important.

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## 2. Problem Statement

During disasters such as floods, earthquakes, cyclones, and landslides, people often face multiple problems simultaneously:

- Lack of timely and reliable information
- Difficulty identifying safe locations
- Blocked or unsafe routes
- Delayed emergency response
- Lack of coordination between citizens and authorities
- Difficulty identifying vulnerable populations
- Limited awareness of shelter capacity and available resources
- Poor connectivity during disasters
- Neglect of livestock and domestic animals during evacuation

Traditional systems often provide information through separate channels, making decision-making slower and less coordinated.

**RELOC8 addresses this gap by creating a unified disaster-response and relocation ecosystem.**

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## 3. Objectives

The major objectives of RELOC8 are:

1. Provide citizens with disaster-related information and risk awareness.
  2. Enable quick communication between citizens and authorities.
  3. Provide a dedicated **Medical SOS** mechanism for emergency medical assistance.
  4. Visualize disaster risks using GIS-based maps.
  5. Identify vulnerable locations and populations.
  6. Support intelligent relocation and evacuation planning.
  7. Provide information about shelters, routes, resources, and safe areas.
  8. Include animals and livestock in disaster-relocation planning.
  9. Maintain essential functionality during poor or unavailable internet connectivity.
  10. Help authorities prioritize emergencies and make faster decisions.
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## 4. System Architecture

RELOC8 follows a modular architecture consisting of:

### **Citizen Layer**

Provides:

- Citizen authentication
- Dashboard
- Risk Map
- Information Exchange
- Medical SOS
- Offline Center
- Donation support
- Animal information
- AI assistance
- Citizen settings/profile

### **Authority Layer**

Provides:

- Authority login
- Command dashboard
- Emergency monitoring
- Medical SOS monitoring
- Alert status management
- Risk and location information
- Response coordination

## GIS & Data Layer

The system is designed to work with:

- Village/location data
- Rainfall information
- River proximity
- Risk levels
- Vulnerability information
- Shelter locations
- Route information
- Capacity/resource information

The planned data architecture uses **PostgreSQL/PostGIS and GeoPandas** for geospatial processing.

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## 5. Progressive Web App & Offline-First Architecture

RELOC8 is designed as a **Progressive Web App (PWA)** so that it can behave more like a mobile application while remaining web-based.

Important features include:

- Web App Manifest
- Service Worker
- Offline asset caching
- Cached application shell
- Offline-first design
- Standalone/full-screen application mode

Static assets such as stylesheets and essential UI resources can be cached so that the application does not simply display a blank screen when connectivity is temporarily lost.

This is particularly important during disasters because network connectivity may become unreliable exactly when the application is needed most.

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## 6. Citizen Dashboard

The Citizen Dashboard provides a central interface for accessing disaster-related services.

It includes a **Welcome and Quick Access section** containing:

- Risk Map
- Information Exchange
- Offline Center
- Donation
- Medical SOS

The interface is designed to minimize unnecessary navigation and make important functions accessible quickly during stressful situations.

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## 7. Medical SOS System

One of the major implemented features of RELOC8 is the **Medical SOS system**.

Citizens can submit an emergency request containing:

- Patient name
- Contact number
- Emergency type
- Current location/landmark
- Description of the emergency

The system performs client-side validation before submitting the request.

After successful submission, the citizen receives immediate confirmation:

**“SOS Sent Successfully.”**

The user is also advised to remain at the current safe location and keep their phone available.

The Medical SOS interface uses a high-visibility emergency design with:

- Red emergency styling
- Warning information
- Heart/medical iconography
- Clear SEND SOS action
- Success confirmation screen

## Emergency Types

The system supports medical situations such as:

- Medical Emergency
- Injury
- Pregnancy Emergency
- Breathing Problem
- Unconscious Person
- Other

These categories can be further aligned with detailed disaster-response triage categories such as critical trauma, maternity emergencies, severe illness/infection, and critical medication shortages.

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## 8. Authority Emergency Dashboard

Submitted Medical SOS requests appear on the **Authority Dashboard**.

Each alert contains structured information such as:

- Patient name
- Contact number
- Emergency type
- Location
- Emergency description
- Request creation time
- Current status

Authorities can manage the emergency lifecycle through:

**PENDING → ACCEPTED/DISPATCHED → RESOLVED**

The dashboard also displays the number of active emergency requests and provides a live indication that the emergency feed is being monitored.

This allows authorities to distinguish between newly received requests, accepted cases, and completed cases.

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## 9. SOS Data Synchronization

The current prototype implements SOS synchronization using browser `localStorage`.

Medical SOS records are stored under:

```
reloc8SOSAlerts
```

The Authority Dashboard periodically checks this stored data using an automatic polling mechanism.

Therefore, a citizen can submit an SOS and the authority dashboard can receive the updated information without manually refreshing the page.

### Current Prototype Limitation

Because this implementation uses browser `localStorage`, synchronization is currently limited to the same browser/storage environment.

For a production deployment, it can be replaced with a cloud backend and real-time communication using technologies such as:

- Firebase/Supabase
- Node.js backend
- PostgreSQL/PostGIS
- WebSockets or real-time subscriptions

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## 10. GIS-Based Risk Mapping

GIS is a core component of RELOC8.

The system is designed to represent disaster risk using multiple geographical and environmental parameters, including:

- Rainfall
- Distance from rivers
- Risk level

- Population vulnerability
- Shelter locations
- Route accessibility
- Available capacity

For the Bihar pilot, example monitored locations include:

- Rampur — Supaul
- Bishanpur — Supaul
- Jorgama — Madhepura
- Pratapganj — Supaul
- Udakishunganj — Madhepura

Example telemetry used in the prototype includes rainfall values and distance from rivers to demonstrate how locations can be classified according to risk.

For example, a location with high rainfall and close proximity to a river can receive a higher disaster-risk classification.

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## 11. Intelligent Relocation System

The central idea of RELOC8 is not only to identify **where the disaster is**, but also to determine:

**“Where should people relocate safely?”**

The proposed decision process is:

**Risk Identification → Vulnerability Assessment → Safe Location Identification → Shelter Capacity → Route Accessibility → Priority Generation → Relocation Recommendation**

The system can consider:

- Current risk
- Population vulnerability
- Distance to safe zones
- Shelter capacity
- Road/route conditions
- Availability of resources
- Emergency severity

This transforms RELOC8 from a simple disaster-reporting application into a **decision-support system for intelligent relocation**.

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## 12. Information Exchange

RELOC8 provides an Information Exchange mechanism through which citizens and authorities can share important disaster-related information.

Examples include:

- Flooded roads
- Blocked routes
- Damaged infrastructure
- Unsafe areas
- Missing persons
- Available resources
- Shelter-related information
- Local emergency conditions

Citizen-generated information can help authorities obtain a more realistic understanding of ground-level conditions.

This creates a two-way communication model:

**Citizen → Authority → Response**

rather than a system where information only flows from authorities to citizens.

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## 13. Offline Center

Disaster situations can cause:

- Internet outages
- Mobile network failures
- Power interruptions
- Temporary connectivity problems

Therefore, RELOC8 includes an **Offline Center** concept for accessing essential information when connectivity is unavailable.

The PWA architecture supports offline availability through service workers and cached assets.

Future versions can extend this functionality to include locally stored:

- Emergency contacts
  - Safety instructions
  - Evacuation procedures
  - Nearby shelter information
  - Previously loaded maps
  - Important disaster guidelines
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## 14. Donation & Relief Support

RELOC8 also provides a Donation section to connect users with relief-support activities.

The objective is to make disaster assistance more organized by connecting available support with affected communities.

Future development can integrate verified NGOs, government relief programs, and transparent resource tracking.

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## 15. Animal & Livestock Assistance

Disaster relocation often focuses primarily on humans while animals and livestock are overlooked.

RELOC8 includes an **Animal Assistance / Information Exchange** component to address this issue.

It can support information related to:

- Missing animals
- Animal rescue
- Livestock relocation
- Animal shelters
- Available veterinary support
- Animal-related emergency information

This makes the relocation ecosystem more inclusive for rural communities where livestock may be an important part of household livelihood.

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# 16. AI-Based Decision Support

AI is intended to assist authorities and citizens in interpreting disaster information.

Potential AI-assisted functions include:

- Risk classification
- Vulnerability scoring
- Emergency prioritization
- Situation summarization
- Shelter recommendations
- Route recommendations
- Resource recommendations
- Identification of high-priority areas

For example, multiple villages can be evaluated based on environmental risk, population vulnerability, emergency requests, shelter availability, and route conditions to generate a priority level.

The AI component is designed as **decision support**, not as a replacement for human authorities.

Final emergency and evacuation decisions remain under authorized human control.

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# 17. Technology Stack

## Frontend

- React.js
- JavaScript
- HTML
- CSS
- Vite
- Lucide React

## Mapping & GIS

- Leaflet
- GeoPandas
- PostGIS

## Data & Storage

- LocalStorage for the current prototype
- PostgreSQL/PostGIS planned for scalable deployment

## **Application Architecture**

- Progressive Web App
- Service Workers
- Offline caching

## **Development Tools**

- VS Code
  - Git
  - GitHub
  - Node.js
  - npm
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# **18. Innovation**

The major innovation of RELOC8 is the integration of multiple disaster-management functions into a single ecosystem.

Instead of having separate systems for maps, emergency reporting, shelters, relocation, and communication, RELOC8 combines them.

## **Key Innovation Areas**

**AI + GIS + Relocation + Emergency Response + Citizen Participation + Offline Support**

The platform focuses not only on disaster monitoring but also on the complete decision-making chain:

**Detect → Assess → Prioritize → Respond → Relocate → Recover**

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# **19. Example Use Case**

Consider a flood-prone village in Supaul.

The system detects that:

- Rainfall is high.
- The village is close to a river.
- The location has a critical risk classification.
- Several vulnerable residents require assistance.
- One or more roads are reported as blocked.
- A nearby shelter has limited capacity.

RELOC8 can combine this information to:

1. Mark the area as high priority.
2. Identify vulnerable residents.
3. Check available shelters.
4. Consider route accessibility.
5. Prioritize emergency requests.
6. Provide authorities with relocation information.
7. Inform citizens about relevant safety information.

This allows authorities to make decisions using multiple sources of information rather than relying on a single risk parameter.

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## 20. Current Implementation Status

The current prototype successfully demonstrates several important workflows.

### Implemented

- React-based application interface
- Citizen dashboard
- Authority dashboard
- Citizen/authority view separation
- Medical SOS form
- Medical SOS validation
- SOS submission
- SOS storage using `localStorage`
- Automatic SOS polling
- Authority SOS monitoring
- Accept/dispatch status
- Resolve status
- Risk-map interface
- Information Exchange interface
- Offline Center interface
- Donation interface
- Animal-related interface

- PWA/offline architecture components
- Git/GitHub development workflow

The **Medical SOS flow has been tested end-to-end:**

**Citizen submits SOS → Alert stored → Authority receives alert → Authority accepts → Alert resolved**

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## 21. Current Limitations

The prototype is currently a demonstration system, so some components can be expanded for production deployment.

**Current limitations include:**

- SOS synchronization currently relies on browser `localStorage`.
- Cross-device real-time communication requires a backend.
- AI decision-making can be expanded with trained models and real datasets.
- GIS analysis can be connected to live government/environmental datasets.
- Real-time GPS tracking is not yet fully integrated.
- Shelter and resource databases require centralized backend integration.
- Production deployment would require authentication, authorization, security, and data protection mechanisms.

These limitations represent the next development stage rather than limitations of the overall concept.

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## 22. Future Scope

RELOC8 can be expanded into a complete disaster-management platform through:

**Real-Time Backend**

- Cloud database
- Real-time alerts
- Multi-user synchronization
- Secure authentication

**Advanced AI**

- Flood prediction
- Risk forecasting
- Population vulnerability prediction
- Automatic emergency prioritization
- Intelligent shelter allocation
- Route optimization

## **Advanced GIS**

- Live rainfall data
- River-level monitoring
- Satellite imagery
- Drone imagery
- Dynamic hazard maps
- Real-time road accessibility

## **Emergency Communication**

- SMS alerts
- Push notifications
- Emergency calling
- Multilingual support
- Voice-based assistance

## **IoT Integration**

Sensors can provide real-time information about:

- River levels
- Rainfall
- Water levels
- Weather conditions
- Infrastructure conditions

## **Inclusive Disaster Management**

Future versions can provide stronger support for:

- Elderly citizens
  - Children
  - Pregnant women
  - Persons with disabilities
  - Medically vulnerable individuals
  - Animals and livestock
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## 23. Social Impact

RELOC8 can help reduce disaster-related losses by improving the speed and quality of decision-making.

Its expected impact includes:

- Faster emergency response
- Better evacuation planning
- Improved risk awareness
- Better use of shelters and resources
- Improved citizen-authority communication
- Inclusion of animals and livestock
- Support during connectivity failures
- More organized disaster relocation

For flood-prone regions such as **Supaul and Madhepura**, such a system can help communities move from reactive disaster response toward **prepared and data-driven relocation planning**.

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## 24. Conclusion

RELOC8 proposes a unified approach to disaster management by combining **GIS, AI-assisted decision support, emergency communication, intelligent relocation, offline functionality, and citizen participation**.

The system connects the complete disaster-response cycle—from identifying risk and receiving citizen alerts to prioritizing emergencies and supporting safe relocation.

The current prototype demonstrates the core interaction between citizens and authorities, particularly through the implemented Medical SOS workflow and authority monitoring system.

With future integration of real-time GIS data, cloud infrastructure, AI models, IoT sensors, GPS, route optimization, and secure communication systems, RELOC8 can evolve into a scalable disaster-management and intelligent relocation platform capable of supporting communities during real-world emergencies.